

SUPERNOVA EXTENDER

Wi-Fi Range Extender & Internet Sharing Device

USER MANUAL

Complete Setup & Feature Guide

For devices already flashed with the SUPERNOVA EXTENDER firmware,
downloaded as a ready-made .bin file from GitHub

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Welcome to SUPERNOVA EXTENDER

SUPERNOVA EXTENDER turns a small, low-cost ESP8266 device into a Wi-Fi range extender. It connects to your existing home or office router and re-broadcasts its own Wi-Fi network, so devices that are too far from the main router can still get a strong signal and internet access.

Once it's up and running, you get a full mobile-friendly control panel — accessible from any phone or laptop browser — where you can:

- Connect the extender to your router (by scanning nearby networks or typing details manually).
- Rename your extender's Wi-Fi network and set its own password.
- Watch live status: signal strength, connected devices, uptime, and more.
- Block specific devices from using your Wi-Fi with a simple firewall.
- Change the admin login, reset the device, and customise the look of the panel.

This manual is written for anyone who has already downloaded a ready-made firmware (.bin) file for their board from GitHub and simply wants to install it and use the device — no programming or Arduino IDE knowledge required.

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What You Need Before You Start

- Your ESP8266 device (already labelled or known as either a 1 MB flash or 4 MB flash board).
- A USB cable to connect the device to your computer for the one-time firmware installation.
- A free flashing tool installed on your computer — such as **NodeMCU PyFlasher**, **ESPHome Flasher**, or the **ESP8266 Flash Download Tool** (all free, one-click tools; see Section 4).
- The correct .bin firmware file for your board, downloaded from the GitHub Releases page (see Section 3).
- A 2.4 GHz Wi-Fi router to connect the extender to (ESP8266 devices cannot use 5 GHz Wi-Fi).

You only need the USB cable and flashing tool once, during installation. After that, everything is managed wirelessly through the web control panel.

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Getting the Firmware File (from GitHub)

The developer publishes ready-to-use .bin files on the project's GitHub **Releases** page. Two versions are provided — pick the one that matches your board.

Your Board	Common Examples	File to Download
1 MB Flash	ESP-01, ESP-01S	SUPERNOVA_EXTENDER_1MB.bin
4 MB Flash	NodeMCU, Wemos D1 Mini, Generic ESP8266 dev board	SUPERNOVA_EXTENDER_4MB.bin

How to check your board's flash size if you're not sure:

- Look for a marking on the chip itself (often printed as 25Q80 = 1 MB, or 25Q32 = 4 MB).
- ESP-01 / ESP-01S modules (small, 8-pin, no USB port) are almost always 1 MB.
- NodeMCU and Wemos D1 Mini boards (larger, with a USB port and more pins) are almost always 4 MB.

Downloading the wrong file for your board can cause the device to fail to boot after flashing. If in doubt, check the seller's product listing or the label on the board.

Steps to download:

- Open the project's GitHub page and click the "Releases" link on the right-hand side.
- Open the latest release at the top of the list.
- Under "Assets", click the .bin file matching your board (1 MB or 4 MB) to download it.

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Installing the Firmware on Your Device

This is the only technical step, and it only needs to be done once. Any of the free tools below will work — pick whichever you find easiest.

Option A — NodeMCU PyFlasher / ESPHome Flasher (recommended, Windows/Mac)

- Download and open the flashing tool.
- Connect your ESP8266 board to the computer with a USB cable.
- Select the correct COM/serial port for your device.
- Click "Choose file" and select the .bin file you downloaded (Section 3).
- Set Baud Rate to 115200 and Flash Mode to "dout" or "qio" (default settings usually work).
- Click "Flash NodeMCU" / "Flash ESP8266" and wait for the progress bar to finish.

Option B — ESP8266 Flash Download Tool

- Open the tool and add your .bin file, setting its start address to 0x00000.
- Select the correct COM port and set the baud rate to 115200.
- Click START and wait for "FINISH" to appear.

Putting the board into flashing mode

Most NodeMCU/Wemos-style boards enter flashing mode automatically over USB. Bare ESP-01/ESP-01S modules usually need GPIO0 held to GND while powering on/pressing reset — if your specific board came with instructions from the seller, follow those.

Once flashing finishes, unplug and re-plug the device (or press its reset button). It will boot straight into SUPERNOVA EXTENDER firmware.

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Turning It On for the First Time

- Power the device using a USB charger or power bank (a USB cable to a computer also works).
- On your phone or laptop, open Wi-Fi settings and look for a new network called **SUPERNOVA_EXTENDER**.
- Connect to it using the default password **12345678**.
- Open a web browser and go to **http://10.10.10.1**
- The SUPERNOVA EXTENDER login page will appear — continue to Section 6.

6 Logging Into the Control Panel

On the login page, enter the default admin credentials and press **Login**:

Field	Default Value
Username	Admin
Password	Password

Change these to something private of your own as soon as you log in — see Section 8.3.

Once logged in, your browser stays signed in for the session. Tap **Logout** on the dashboard whenever you want to end the session manually.

7 Dashboard Tour

After logging in, you'll see nine buttons arranged in a grid. Each opens a different part of the control panel:

Button	What it does
Contact	Opens the developer's support/Facebook page
Sound	Turns the panel's click/beep sounds on or off
Theme	Cycles through 15 colourful background themes
Settings	Wi-Fi name/password, device IP, admin login, factory reset
Status	Live view of your connection, signal, and connected devices
Manual	Type in a router name/password to connect to it directly
Scan	Lists nearby Wi-Fi networks so you can connect with one tap
Firewall	Block or allow specific devices on your network
Logout	Signs you out of the control panel

A closer look at every screen inside the control panel.

8.1 Settings — Wi-Fi Name, Password & Mode

- **Wi-Fi Name (AP Name):** the network name your phone/laptop sees and connects to.
- **Wi-Fi Password (AP Password):** needed to join the extender's network (at least 8 characters).
- **Network Mode:** choose **Password Mode** for a secured network, or **Public** for an open network with no password.
- Tap **Save AP Settings** to apply. The extender restarts its Wi-Fi immediately with the new name/password — reconnect your devices to the new network name afterward.

8.2 Settings — Device IP Address

This is the web address you type into your browser to reach the control panel (default 10.10.10.1). Change it here if it clashes with another device on your network. The extender restarts after saving, so browse to the new address afterward.

8.3 Settings — Admin Login Details

Sets the username and password used to log into this control panel (different from the Wi-Fi password). Fill in both fields and tap **Save Admin Credentials**. Do this early — it stops other people on your Wi-Fi from getting into your settings.

8.4 Settings — Factory Reset

Tap **Reset All Settings to Default** to erase everything you've configured (Wi-Fi details, admin login, firewall rules) and return the device to its out-of-the-box state. You'll be asked to confirm first, since this cannot be undone.

8.5 Status — Live Network Information

This screen updates itself automatically every few seconds and shows:

- Whether the extender is connected to your router, and its Wi-Fi signal strength.
- How many devices are currently connected to the extender's own Wi-Fi.
- How long the device has been running since it was last powered on.
- The current time (synced automatically over the internet).
- A list of every connected device, along with how long each has been connected.

8.6 Manual Connect

Use this if you'd rather type your router's name and password directly instead of scanning (handy for hidden networks). Enter the details and tap **Connect** — the panel will let you know once the connection succeeds or if it fails.

8.7 Scan — Find Nearby Wi-Fi

Shows every nearby 2.4 GHz Wi-Fi network with its signal strength and security type. Tap **Set** next to your router's name to jump straight into Manual Connect with the name already filled in — just add the password and tap Connect.

■ 8.8 Firewall — Block Unwanted Devices

Keep unwanted devices off your Wi-Fi by blocking them using their MAC address (a unique code every device has, usually found in that device's own Wi-Fi settings).

- Type the MAC address (format AA:BB:CC:DD:EE:FF) and tap **Block MAC Address**.
- Blocked devices are disconnected automatically within a few seconds and kept off the network.
- Up to 5 devices can be blocked at the same time.
- Tap **Unblock** next to any entry to allow that device back on.

■ 8.9 Theme, Sound, Contact & Logout

- **Theme:** instantly changes the panel's background colour scheme — purely visual.
- **Sound:** toggles short confirmation tones for button taps.
- **Contact:** opens the developer's page for help or updates.
- **Logout:** signs you out and returns to the login screen.

9 The Physical Reset Button

Every board has a small physical button (usually labelled "Flash" or "Boot"). Press and hold it for **5 seconds** to instantly erase all your settings and return the device to its factory defaults — useful if you're ever locked out.

This only works as a physical button press — it cannot be triggered from the web control panel, which keeps it safe from being reset accidentally over Wi-Fi.

10 Understanding the Status LED

The small blue light on the device tells you its connection state without needing to open the control panel:

LED Behaviour	What It Means
Solid ON (steady light)	Connected to your router — internet sharing is active
Fast blinking	Trying to connect (or reconnect) to your router
Slow blinking	Standalone mode — not linked to any router yet
Very fast strobing	The reset button is currently being held down

11 How Internet Sharing Works

Once the extender connects to your router, it automatically shares that internet connection with every device that joins its own Wi-Fi network — just like a normal Wi-Fi repeater or hotspot. You don't need to turn this on manually; it happens as soon as the LED turns solid.

- If your router restarts or the extender loses the connection, it automatically tries to reconnect every 30 seconds — no action needed from you.
- Internet sharing resumes automatically the moment the connection is restored.

12 Default Login Details & Security Tips

Setting	Default Value
Wi-Fi Network Name	SUPERNOVA_EXTENDER
Wi-Fi Password	12345678
Device Address	10.10.10.1
Admin Username	Admin
Admin Password	Password

Recommended first steps after setup:

- Change the admin username/password (Section 8.3) so only you can access the settings.
- Change the Wi-Fi password (Section 8.1) to something private, unless you intend to run an open network on purpose.
- Use the Firewall (Section 8.8) to remove any device you don't recognise from your network.

13 Troubleshooting

Problem	What to Try
Can't find the SUPERNOVA_EXTENDER Wi-Fi network	Make sure the device is powered on; wait 15-20 seconds after power-up
Can't open 10.10.10.1 in the browser	Confirm your phone/laptop is connected to the SUPERNOVA_EXTENDER Wi-Fi network, not the router
LED keeps fast-blinking and never turns solid	Your router password may be wrong, or the router is out of range — try Scan or Manual Connect
Connected devices have Wi-Fi but no internet	Check the Status screen to confirm the extender itself is connected to your router
Forgot the admin password	Hold the physical reset button for 5 seconds, then log in again with the defaults in Section 12
Device won't turn on / no LED at all after flashing	Re-check that you flashed the .bin file matching your board's flash size (Section 3)
Scan shows no networks at all	Make sure your router broadcasts a 2.4 GHz network (5 GHz-only networks won't appear)

14 Quick Reference Card

Keep this page handy — everything you need at a glance.

Item	Detail
Default Wi-Fi Name	SUPERNOVA_EXTENDER
Default Wi-Fi Password	12345678
Control Panel Address	http://10.10.10.1
Default Admin Login	Admin / Password
Factory Reset	Hold the physical button for 5 seconds
LED - Solid ON	Connected, internet sharing active
LED - Fast Blink	Trying to connect to router
LED - Slow Blink	Standalone mode, no router linked
LED - Rapid Strobe	Reset button is being held down
Auto-Reconnect Interval	Every 30 seconds if disconnected
Max Blocked Devices	5 MAC addresses

SUPERNOVA EXTENDER — User Manual

This manual documents the device's control panel and everyday usage for end users.